

Insurance Cost Analysis and Prediction

Capstone Project – Data Science

Sharvin Kubal

March 2026

1. Introduction

This project focuses on analyzing an insurance dataset to understand the factors influencing medical insurance charges. The goal is to explore patterns in the data and develop a predictive model to estimate insurance costs using machine learning techniques.

2. Problem Statement

To analyze the key factors affecting insurance charges and build a regression model capable of predicting medical expenses based on individual attributes.

3. Dataset Description

The dataset consists of the following features:

- Age – Age of the policyholder
 - Sex – Gender of the individual
 - BMI – Body Mass Index
 - Children – Number of dependents
 - Smoker – Smoking status
 - Region – Residential region
 - Charges – Medical insurance cost (Target Variable)
-

4. Data Preprocessing

- Checked for missing values (none found)
- Removed duplicate entries

- Converted categorical variables using one-hot encoding
 - Prepared feature matrix and target variable
-

5. Exploratory Data Analysis

- Distribution of insurance charges was right-skewed
 - Smokers were found to incur significantly higher charges
 - Age showed a positive relationship with insurance cost
 - BMI had moderate influence
 - Gender and region showed minimal impact
 - Correlation heatmap used to identify relationships
-

6. Model Building

A Linear Regression model was implemented.

Steps performed:

- Dataset split into 80% training and 20% testing
 - Model trained using training data
 - Predictions generated on test data
-

7. Model Evaluation

Evaluation Metrics Used:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- R^2 Score

The model achieved an R^2 score of 0.80, indicating strong predictive performance and good explanatory power.

8. Conclusion

The analysis identified smoking as the most influential factor affecting insurance charges, followed by age and BMI. The Linear Regression model successfully predicts insurance costs with good accuracy. This project demonstrates practical implementation of data preprocessing, exploratory analysis, and regression modeling techniques.